

Mathematical Symbols (Public Domain / Unicode)

1. Basic Arithmetic

- + Addition
- − Subtraction
- × Multiplication
- ÷ Division
- = Equals
- ≠ Not equal to
- ≈ Approximately equal
- ≡ Identically equal
- ∝ Proportional to

2. Algebra

- √ Square root
- ∛ Cube root
- ∜ Fourth root
- ∞ Infinity
- | Divides
- ∤ Does not divide
- ± Plus-or-minus
- ∓ Minus-or-plus

3. Sets

- ∈ Element of
- ∉ Not an element of
- ⇒ Contains as member
- ∅ Empty set
- ⊂ Proper subset
- ⊃ Proper superset
- ⊆ Subset or equal
- ⊇ Superset or equal
- ∪ Union
- ∩ Intersection
- ⊔ Disjoint union
- ⊕ Symmetric difference

4. Logic

- ¬ Not
- ∧ Logical AND
- ∨ Logical OR
- ⊕ Exclusive OR

- $\Rightarrow$ Implies
- $\Leftrightarrow$ If and only if
- $\forall$ For all
- $\exists$ There exists
- $\nexists$ There does not exist
- $\vdash$ Entails
- $\models$ Models

5. Calculus

- ∂ Partial derivative
- ∇ Nabla / gradient
- $\int$ Integral
- $\iint$ Double integral
- $\iiint$ Triple integral
- $\oint$ Contour integral
- $\sum$ Summation
- $\prod$ Product

6. Geometry

- $\sphericalangle$ Angle
- $\sphericalangle^{\circ}$ Measured angle
- $\perp$ Perpendicular
- $\parallel$ Parallel
- $\triangle$ Triangle
- $\square$ Square
- $\oslash$ Diameter symbol

7. Number Sets

- $\mathbb{N}$ Natural numbers
- $\mathbb{Z}$ Integers
- $\mathbb{Q}$ Rational numbers
- $\mathbb{R}$ Real numbers
- $\mathbb{C}$ Complex numbers
- $\mathbb{P}$ Prime numbers

8. Relations

- $<$ Less than
- $\leq$ Less than or equal
- $>$ Greater than
- $\geq$ Greater than or equal
- $\ll$ Much less than
- $\gg$ Much greater than
- $\sim$ Similar to

- $\cong$ Congruent to
- $\approx$ Approximately equal

9. Probability & Statistics

- $P(A)$ Probability of event A
- $E[X]$ Expected value
- $\text{Var}(X)$ Variance
- σ Standard deviation
- μ Mean
- ρ Correlation
- $\sim$ Distributed as
- $\propto$ Proportional to

10. Miscellaneous Technical Symbols

- $\top$ Top (truth)
- $\perp$ Bottom (falsehood)
- $\mapsto$ Maps to
- $\rightarrow$ Function arrow
- $\leftrightarrow$ Bidirectional arrow
- $\Uparrow$ Upwards double arrow
- $\Downarrow$ Downwards double arrow